AI ENGINEERING / APPLIED AI / CASE STUDY

PROOF AI

CASE STUDY

04

Proof AI tests how far independent checks and abstention can stop wrong automatic acceptance.

PROPOSED RESEARCH / NOT YET VERIFIED

THE BUSINESS CONTEXT

A finance operations team receives supplier invoices and matching purchase documents as scans and PDFs. A model can read them quickly but sometimes reads them wrongly with full confidence. The workflow matters because an accepted error becomes a payment. This case study uses synthetic documents only.

WHO IS INVOLVED AND WHAT THEY NEED

01	**Accounts payable lead**	Wants fewer manual keying tasks without accepting silent errors.
02	**Finance controller**	Needs every acceptance to be explainable and replayable at audit.
03	**ML engineer**	Needs calibrated confidence and labelled failures to improve extraction.

SCOPE AND HONEST LIMITS

- ⚠ Synthetic and openly licensed documents only. No production data is used.
- ⚠ Rule consistency does not prove that a value is factually correct.
- ⚠ No claim of compliance certification or measured savings is made.
- ⚠ Results describe the labelled set and may not transfer to other layouts.

CURRENT WORKFLOW AND FAILURE POINTS

04

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HOW THE WORK RUNS TODAY

1 TODAY
Receive documents
Scans and PDFs arrive by email and shared folders with no common structure.

→

2 TODAY
Key fields by hand
A clerk types supplier, totals, dates and references into the ledger.

→

3 TODAY
Match to order
The clerk searches for the order and receipt and compares figures by eye.

↓

4 TODAY
Query mismatches
Differences go to colleagues by email with no shared record.

→

5 TODAY
Approve payment
An approver signs off from a summary that hides the source evidence.

→

6 TODAY
Archive
Documents are filed and the reasoning behind each decision is lost.

WHERE IT BREAKS

Slow manual keying	Repeated typing consumes time and introduces its own transcription errors.
Weak evidence trail	Why a document was accepted cannot be reconstructed later.
Unchecked model output	A fluent extraction can be wrong while sounding certain.
Inconsistent review	Different reviewers apply different standards to the same mismatch.

REDESIGNED WORKFLOW AND DECISION AUTHORITY

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THE REDESIGNED WORKFLOW

1 SYSTEM

Ingest and version

Each document is stored with a content hash and a source reference.

2 SYSTEM

Extract to a schema

A local model proposes typed fields with a confidence score for each.

3 SYSTEM

Run deterministic rules

Arithmetic, duplicate, identity and cross document checks record explicit outcomes.

4 SYSTEM

Route by threshold

Low confidence or any failed rule abstains and sends the case to a reviewer.

5 PERSON

Review with evidence

A reviewer sees source regions beside extracted values and decides.

6 PERSON

Approve and replay

A named approver releases the case and the full decision can be replayed.

SYSTEM STEP PERSON DECISION

WHERE AUTHORITY STAYS WITH A PERSON

Accept an extraction as correct	ACCOUNTS PAYABLE LEAD	The system may propose but a person owns every accepted figure.
Set abstention thresholds	FINANCE CONTROLLER	Thresholds trade throughput against risk and need a named owner.
Approve payment release	AUTHORISED APPROVER	Payment is consequential and stays outside any automated path.

ARCHITECTURE, DATA AND CONTROLS

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SYSTEM ARCHITECTURE

1	**Document intake**	Versioned storage with hashes so every input is traceable to its source.
2	**Extraction**	Local OCR and compact models emitting typed fields with confidence.
3	**Rule verification**	Deterministic checks for totals, duplicates, identity and cross document links.
4	**Decision and review**	Routing, reviewer interface and persistent state with provenance.
5	**Evaluation**	Labelled dataset, error taxonomy and calibration reporting.

CORE DATA ENTITIES

- **Document**
- **Extracted field**
- **Confidence score**
- **Rule outcome**
- **Review case**
- **Decision record**

RISKS AND THE CONTROL FOR EACH

Confident but wrong extraction →	Independent rules and abstention thresholds stop acceptance without review.
Coherent but wrong interpretation →	Sample accepted cases for human audit and track the undetected error rate.
Duplicate invoice accepted →	Hash and field level duplicate checks run before any approval step.
Threshold drift over time →	Recalibrate against fresh labelled samples and record each threshold change.

EVALUATION AND DELIVERY PLAN

04

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HOW SUCCESS IS MEASURED

False automatic accept rate	Share of wrong documents accepted without review on the labelled set.
Human review volume	Share of documents routed to a reviewer at each threshold setting.
Extraction precision and recall	Field level agreement with labels, reported per field type.
Confidence calibration	Gap between stated confidence and observed accuracy across bins.
Undetected consistent errors	Wrong values that pass every rule, counted and described in the report.

TEST PLAN

1. **Model only, rules only and hybrid baselines run on the same labelled synthetic set.**
2. **Inject duplicates, missing fields, OCR noise and total conflicts at known rates.**
3. **Replay any decision from stored provenance and confirm the same outcome.**
4. **Sweep abstention thresholds and plot errors against review volume.**

DELIVERY PLAN

01 ALIGN	02 UNDERSTAND	03 DESIGN	04 DELIVER	05 LAUNCH
Agree the document types, the cost of a wrong acceptance and who owns thresholds.	Map the current manual route and sample the error types people see today.	Define schemas, rule set, review states and the evaluation protocol.	Build the pipeline and reviewer interface and run the baseline comparison.	Run in shadow beside the manual process and compare outcomes before any reliance.
DECISION Which errors are unacceptable	**DECISION** Which failure classes to model	**DECISION** Approve rules and thresholds method	**DECISION** Accept the evaluation evidence	**DECISION** Whether to extend beyond shadow

DELIVERABLES

- **Versioned labelled evaluation dataset**
- **Rule library with documented outcomes**
- **Decision replay and review interface**
- **Calibration and failure analysis report**
- **Threshold selection guide for owners**